Jason Mac

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Education

University of British Columbia — Vancouver, BC, Canada
Bachelor of Applied Science in Computer Engineering — *Expected Graduation: May 2028*

- GPA: 88.6 % - Dean's List
- Coursework: Advanced Algorithms, Database Systems, Computer Architecture, Machine Learning, Probability

Douglas College — New Westminster, BC, Canada
Engineering Associate, GPA: 4.24/4.33 — *May 2024*

Work Experience

Tenstorrent — Toronto, ON, Canada
Software Engineer Intern, AI Inference — *May 2026 – Present*

- Contributed to a C++ disaggregated LLM inference engine and the tt-mlir compiler stack for PyTorch/JAX
- Deployed Tenstorrent's first disaggregated speculative decoding setup for Kimi K2.7 with a DFlash draft model on 20 decode and 8 prefill nodes via NVIDIA Dynamo's router, hardening it for its production launch on OpenRouter
- Implemented engine-side multi-token prediction (MTP) speculative decoding for models like DeepSeek and GLM to speed up token generation, defining an interface with kernel developers for lookahead tokens in MTP prefill
- Designed a feature enabling prefix caching for linear attention models like Kimi K3, migrating recurrent state snapshots across device memory, host memory, and a global key-value store to reduce TTFT on shared prefixes
- Developed and extended multi-host distributed runtime systems, using Slurm to orchestrate multi-node jobs and integrating performance profiling tools

Projects

Real-Time Chat Application | *Rust, Axum, React, TypeScript, PostgreSQL* — GitHub

- Built a full-stack real-time chat application with a Rust backend using Axum and a React TypeScript frontend
- Implemented WebSocket connections for real-time bidirectional messaging using Tokio and a shared users map
- Designed a RESTful API with JWT-based authentication and password hashing via Argon2
- Built a responsive single-page frontend with conversation management, user search, and real-time message delivery

Lua Bytecode Compiler & Virtual Machine | *C++, Lua* — GitHub

- Built a bytecode compiler and stack-based virtual machine for a subset of Lua 5.1 in C++
- Implemented a recursive descent parser producing an AST, converted to bytecode via the visitor pattern
- Designed a bytecode instruction set with support for control flow, local/global variables, and first-class functions
- Implemented lexical scoping, closures, call frames, and native C++ function bindings in the VM runtime

UBC Course Planner | *C#, ASP.NET, .NET MAUI, PostgreSQL* — GitHub

- Developed a cross-platform course planning application using .NET MAUI and the MVVM architecture
- Designed a RESTful backend using ASP.NET Core and Entity Framework Core to serve course data
- Implemented ViewModels that asynchronously fetch live course and offering data for the GUI
- Developed an interactive calendar page allowing users to visualize and drag-and-drop course offerings

Tron Light-Cycle Game | *C, RISC-V, Altera FPGA* — GitHub

- Developed a two-player Tron-style light-cycle game in C on a RISC-V softcore running on FPGA hardware
- Rendered graphics directly to a memory-mapped VGA framebuffer, including borders, obstacles, and player trails
- Used timer-based interrupts with inline RISC-V assembly to drive game updates
- Handled player input using FPGA KEY-based hardware interrupts, coordinating multiple ISRs via shared state

Technical Skills

Languages: C++, C, Python, TypeScript, Rust, SQL
Tools: Linux, Git, GitHub Actions, CMake, Docker, Kubernetes, Slurm
Libraries/Frameworks: React, PyTest, PyTorch, CUDA, MPI